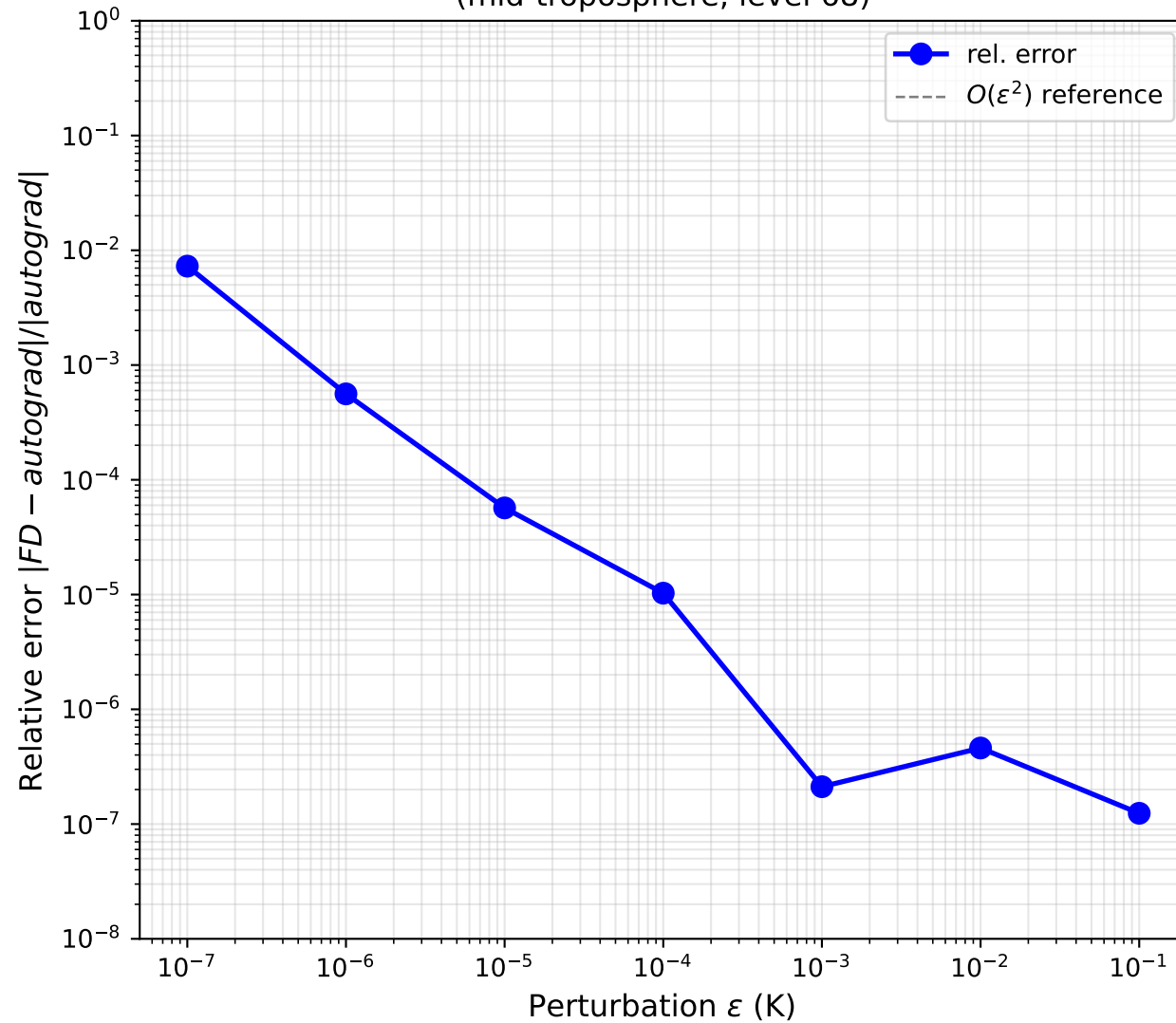
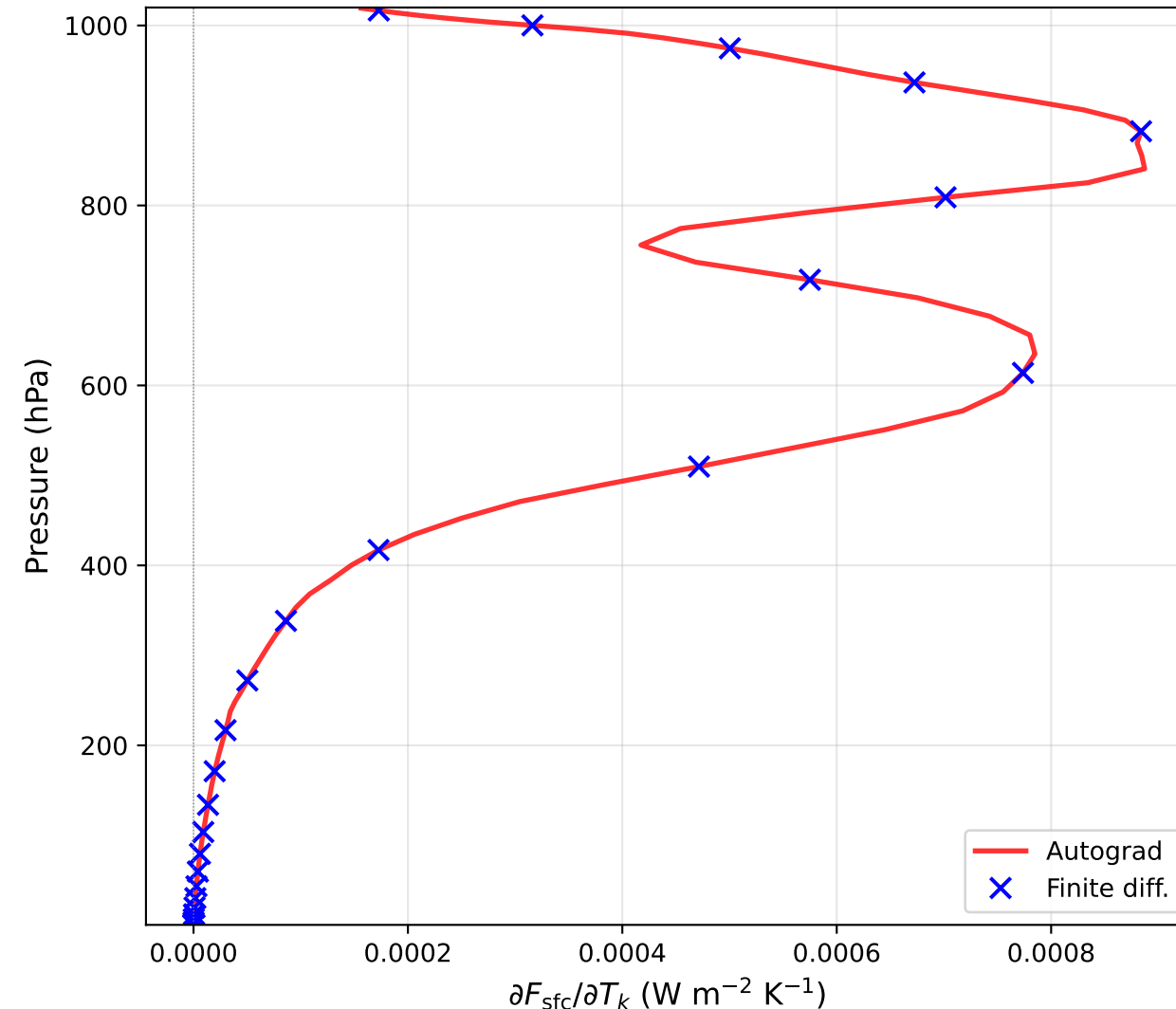


Gradient Verification: Autograd vs Central Finite Differences (Clear-sky SW surface flux, ERA5 column, $\mu_0 = 0.5$, 137 levels)

(a) FD Convergence: $\partial F_{\text{sfc}}/\partial T_k$
(mid-troposphere, level 68)



(b) Vertical Gradient Profile
 $\partial F_{\text{sfc}}/\partial T_k$ at each level



(c) FD Convergence: $\partial F_{\text{sfc}}/\partial \mu_0$
(autograd = 1268.2 W m $^{-2}$)

